

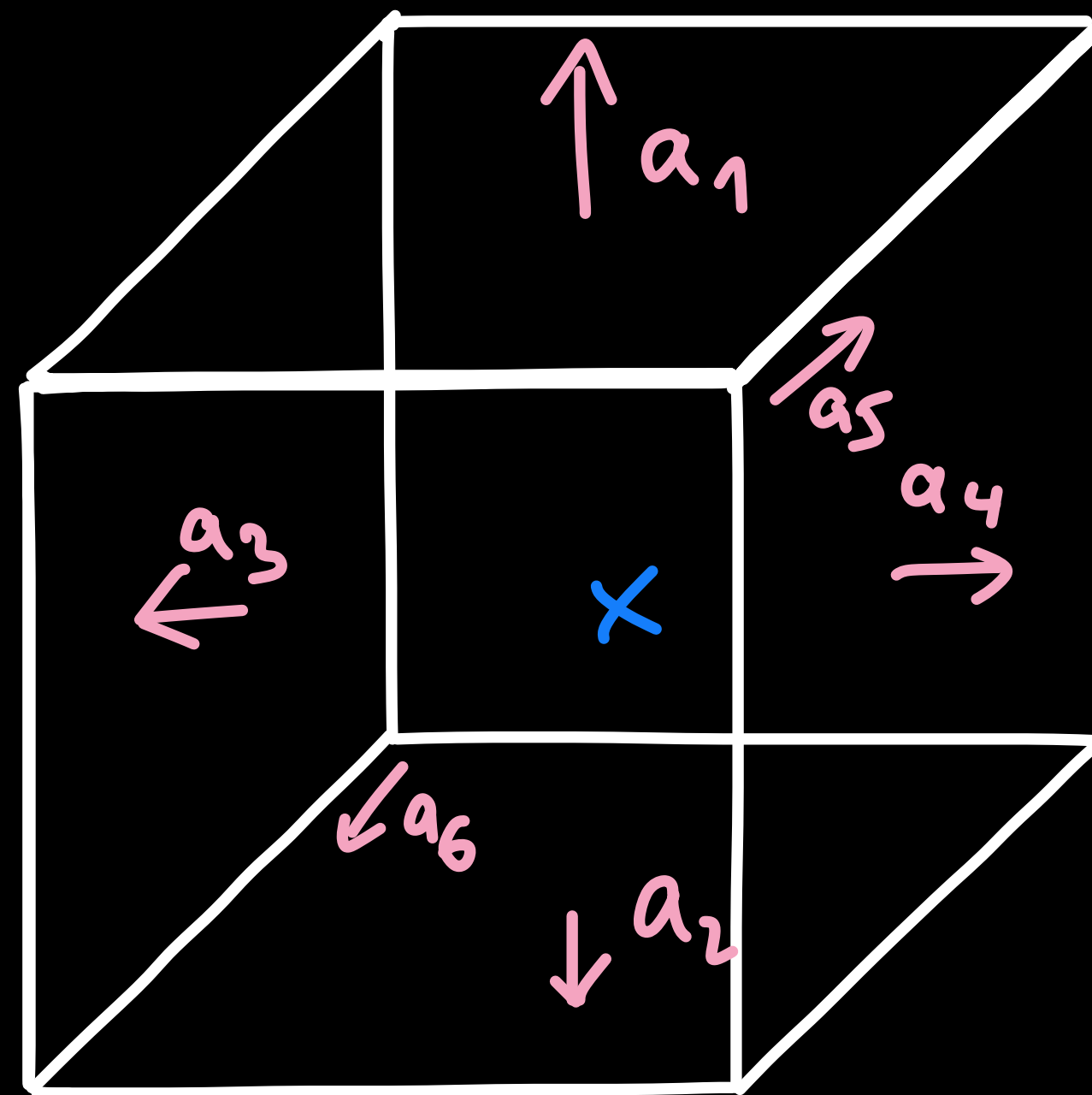
On the complexity of the simplex method

Eleon Bach

maximize $c^T x$

subject to $Ax \leq b$

$$A = \begin{pmatrix} -a_1 & - \\ \vdots & \\ -a_n & - \end{pmatrix}$$





38. Mathematical Tables Project computers with adding machines

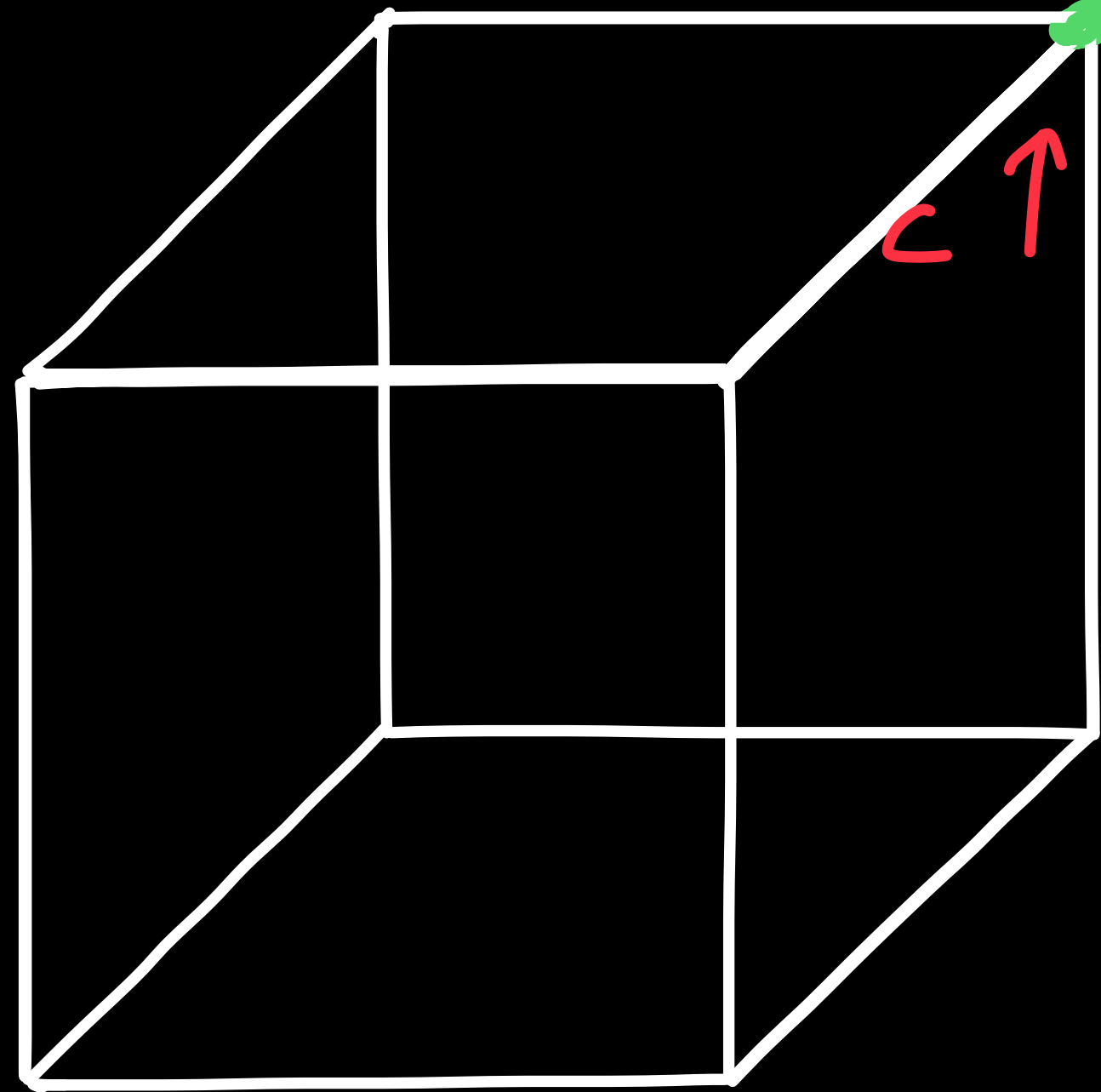
David A. Grier, "When computers were human."

maximize $c^T x$

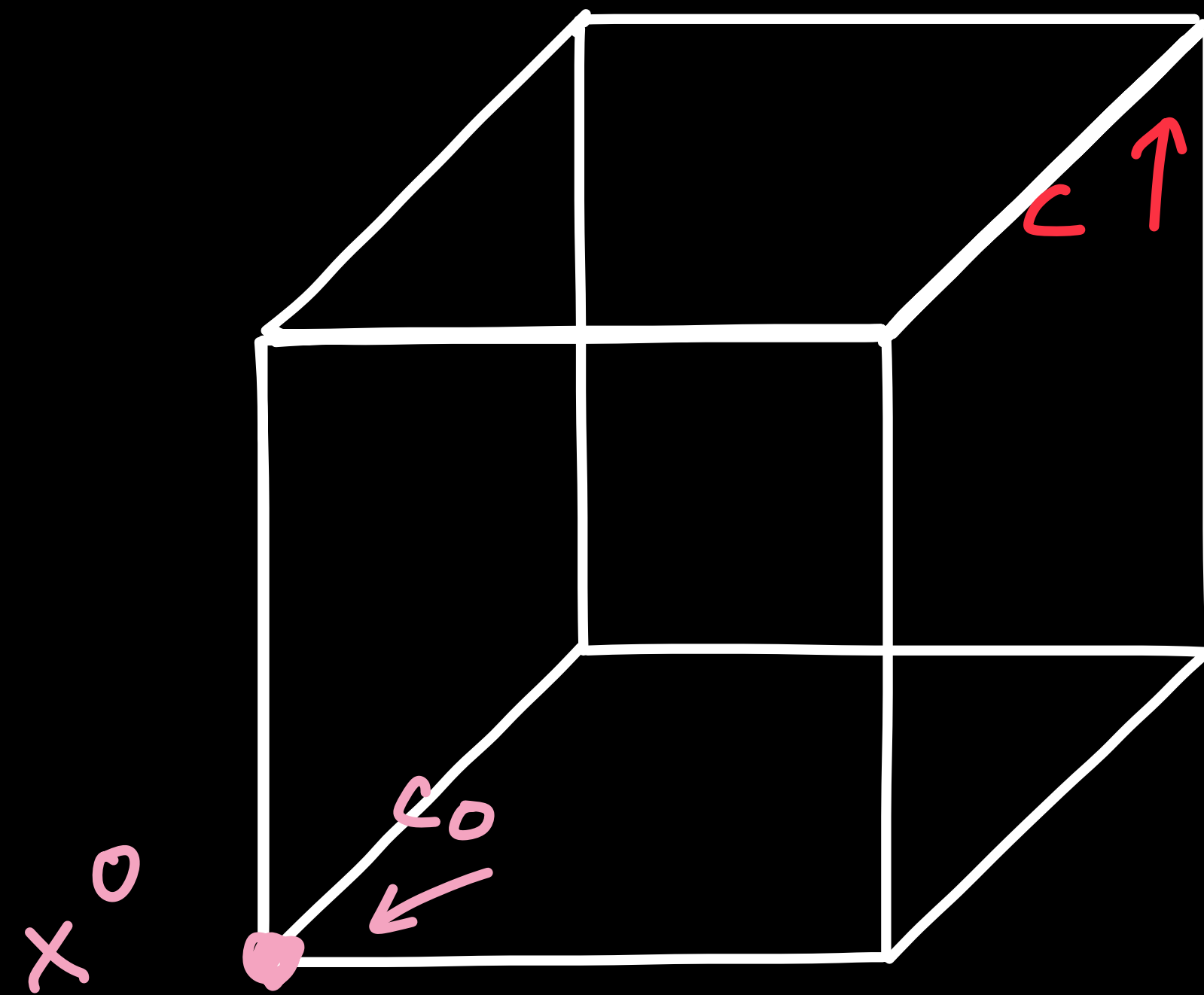
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$$A = \begin{pmatrix} -a_1 & - \\ \vdots & \\ -a_n & - \end{pmatrix}$$

$$\text{opt} = \max_{\{x: Ax \leq b\}} c^T x$$

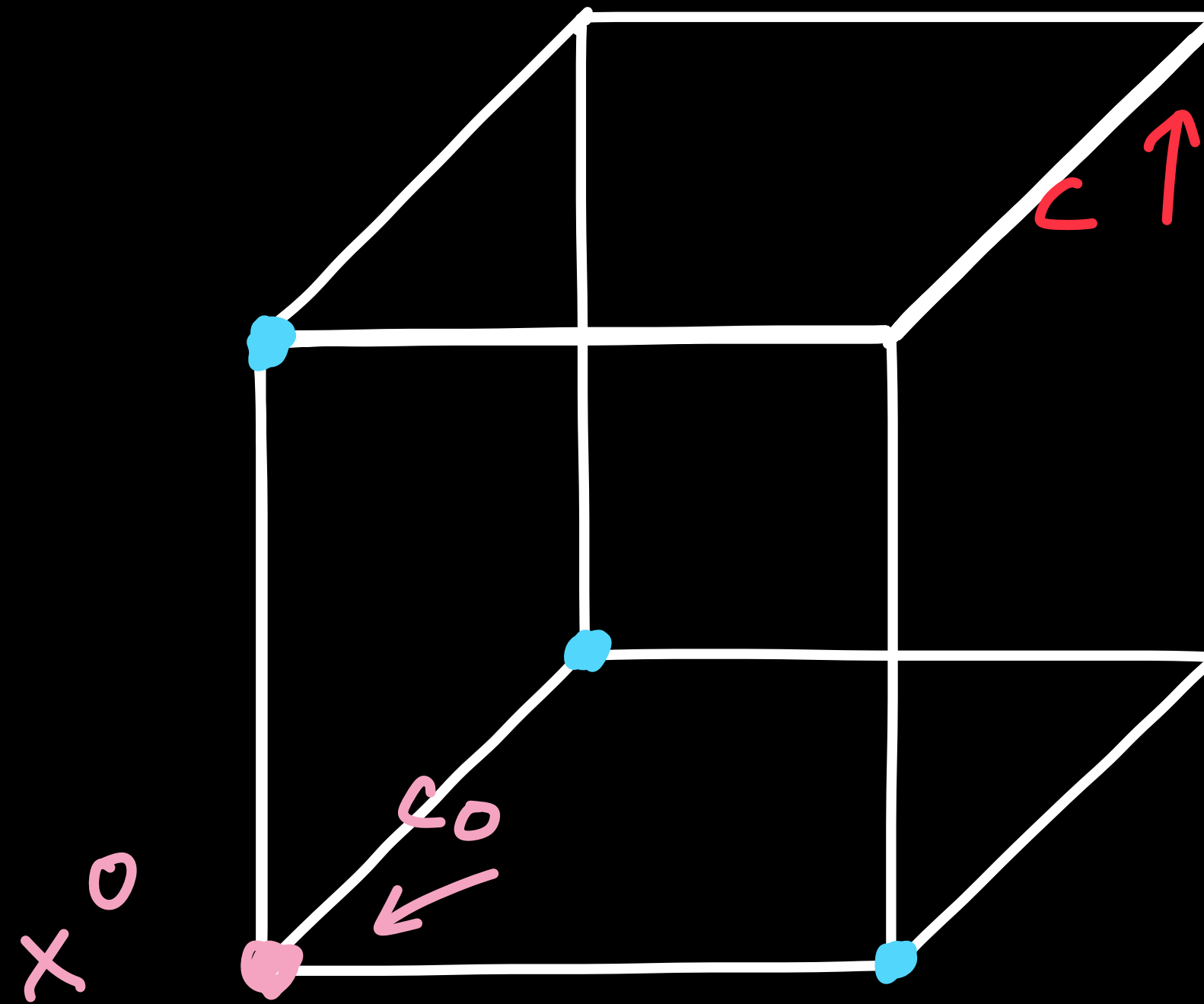


Simplex method



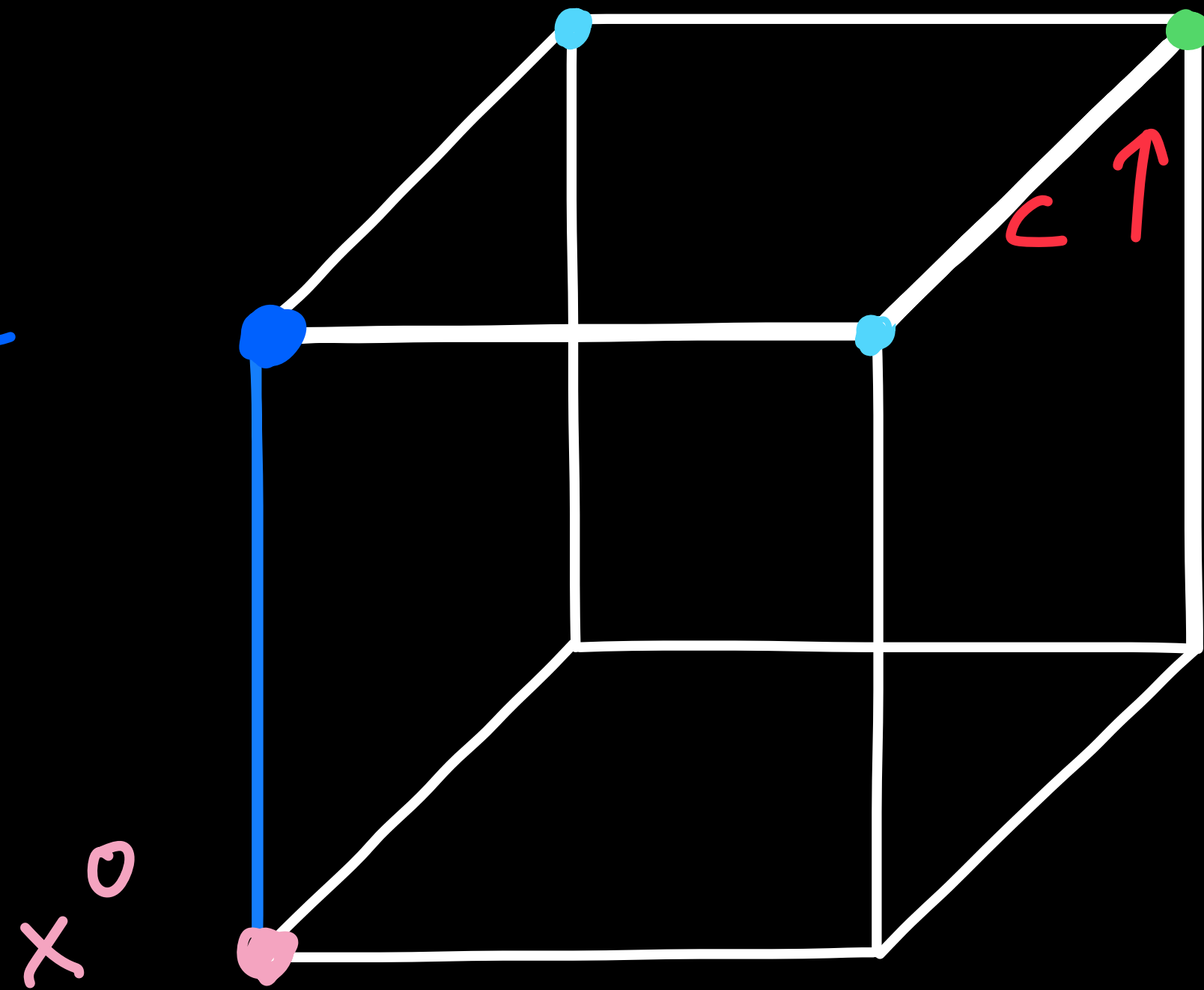
Phase 1

Simplex method



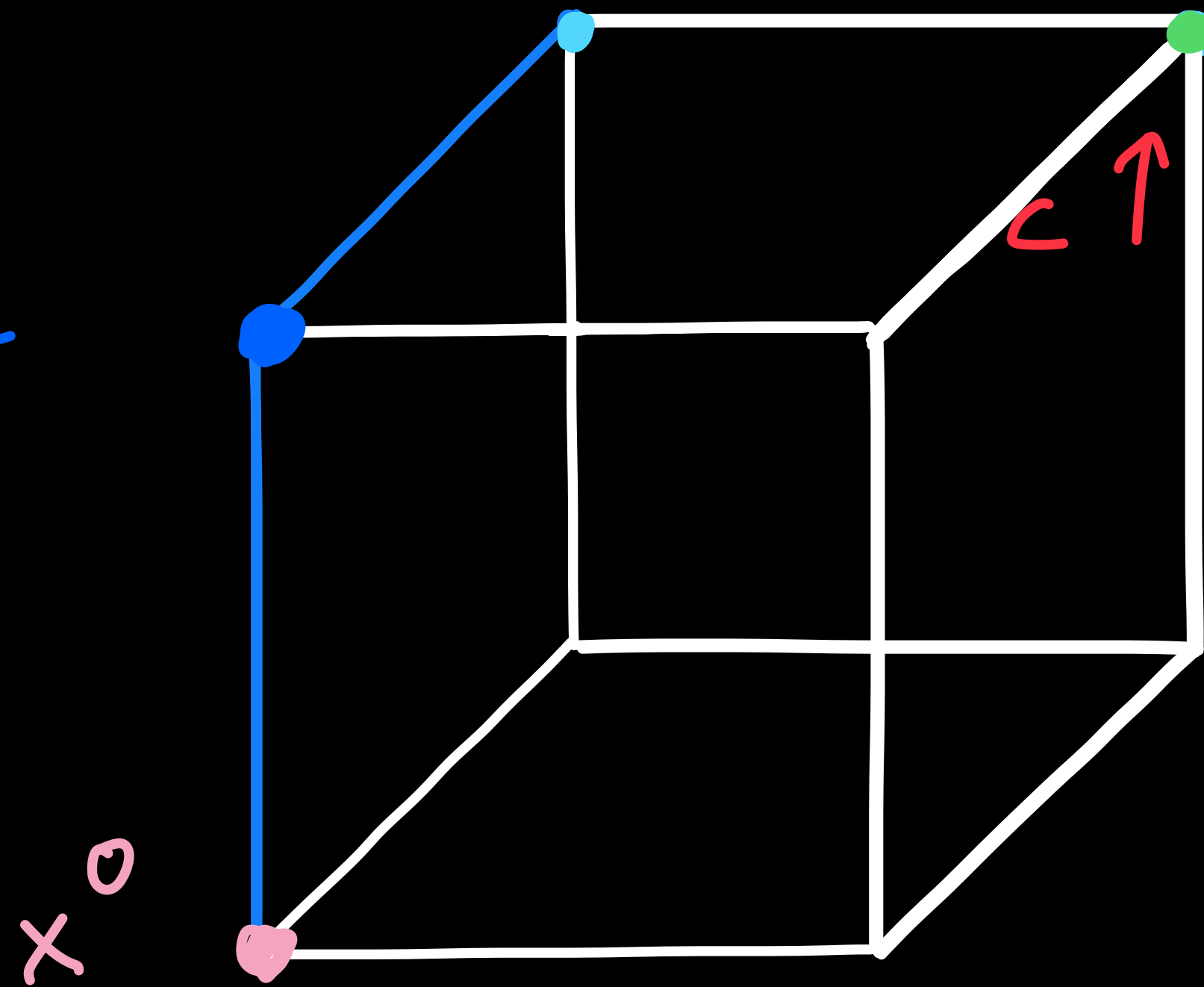
Simplex method

pivot rule



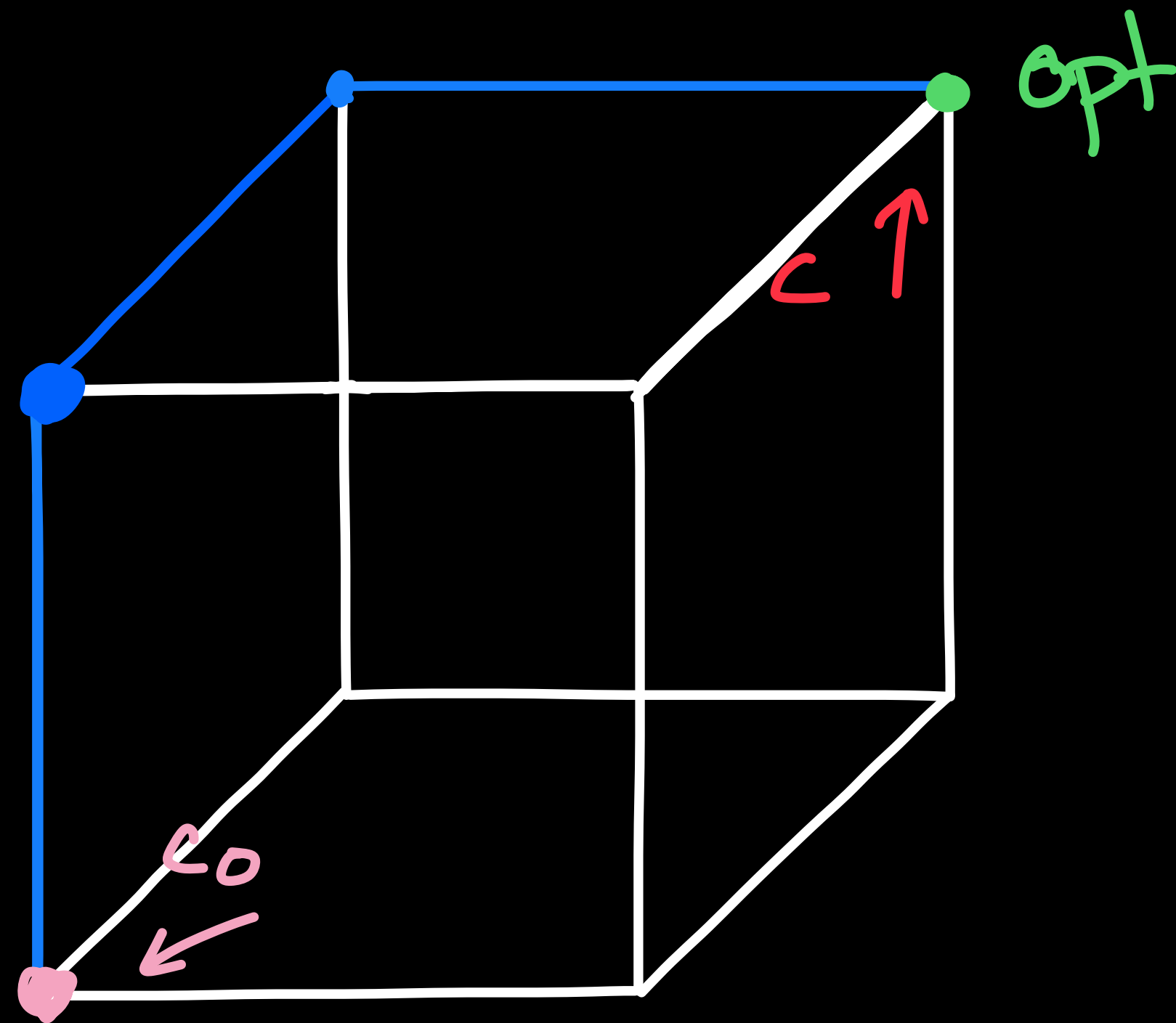
Simplex method

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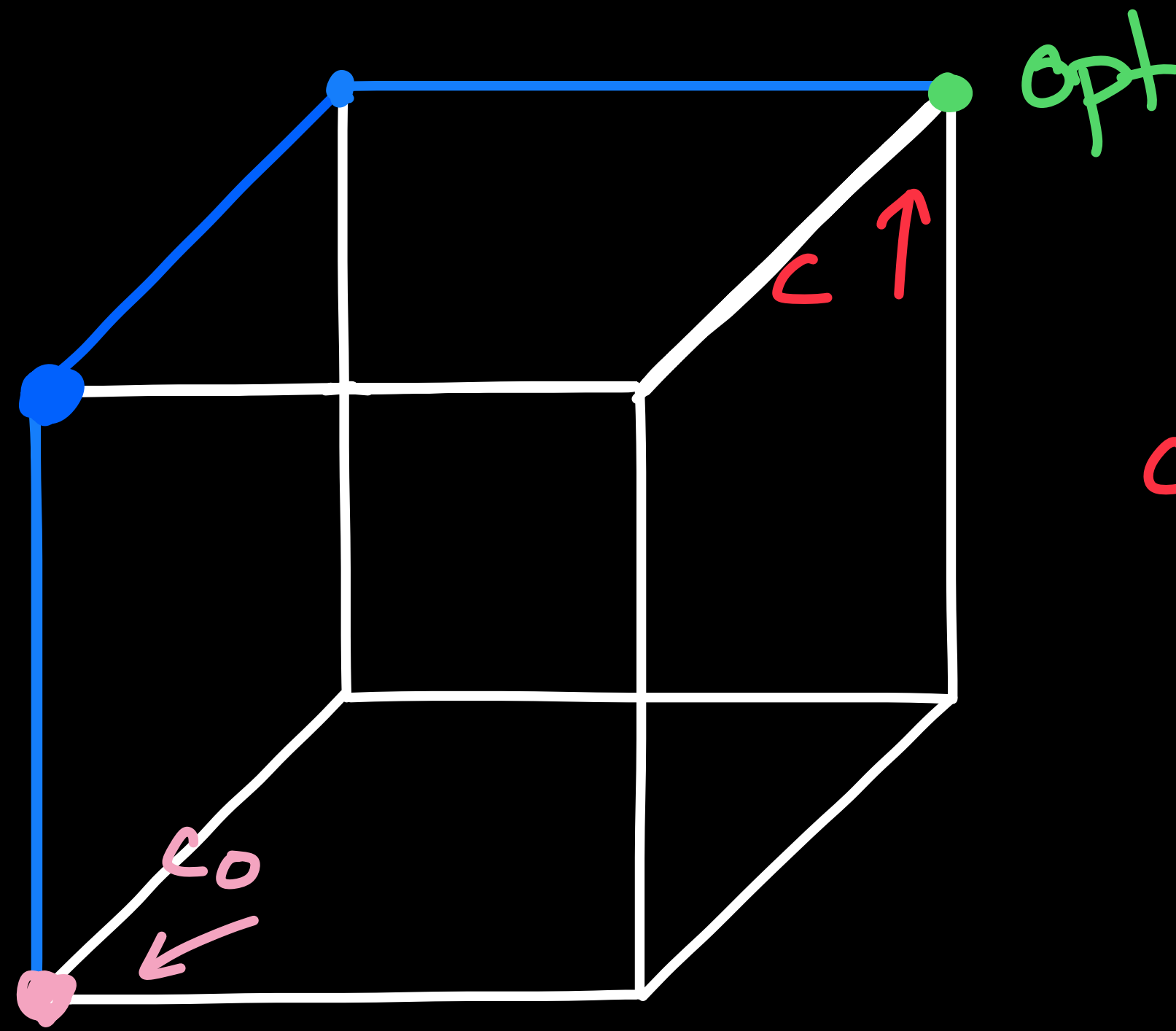
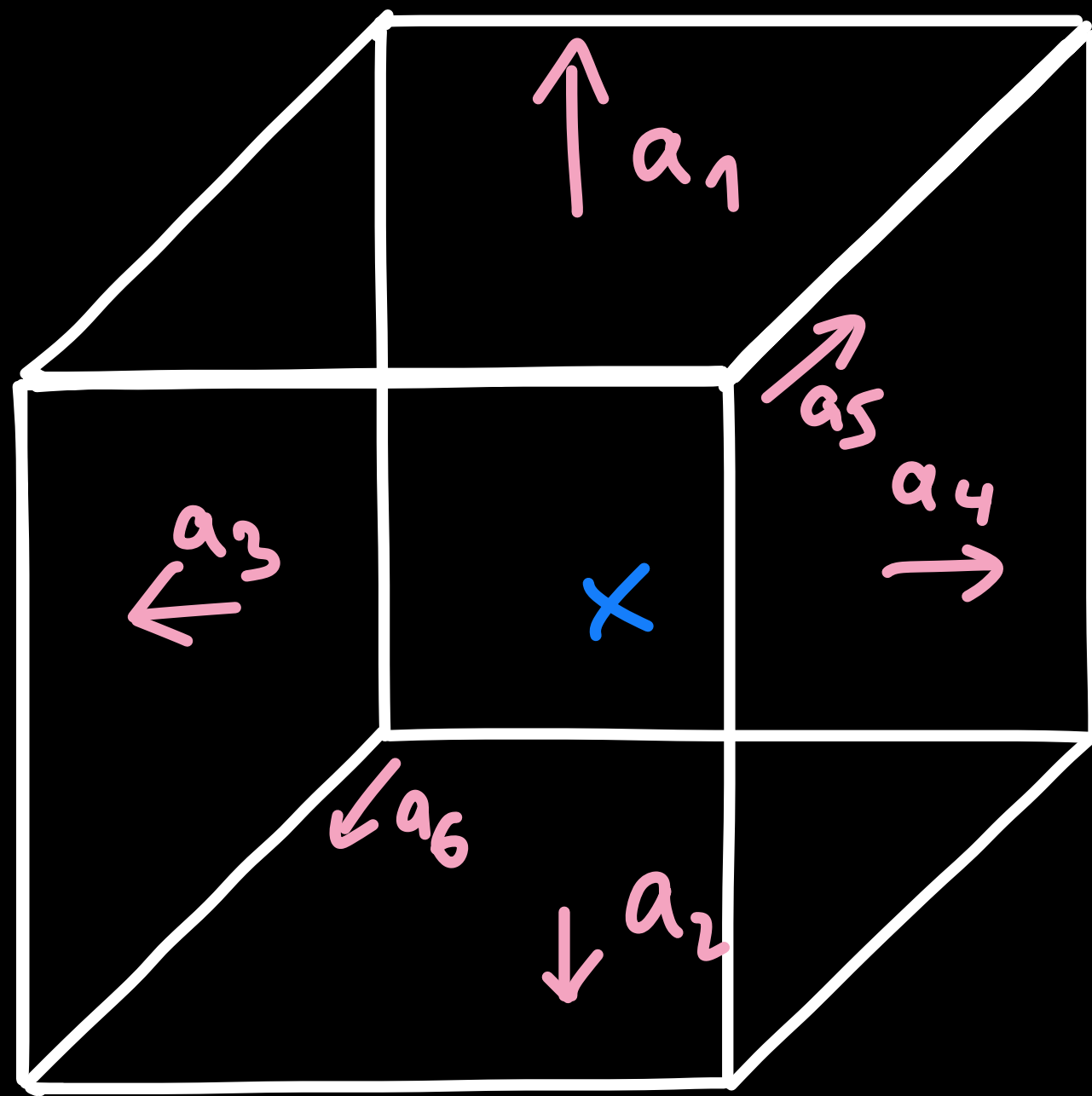


Simplex method

pivot rule



Simplex method



$$c \in \text{cone}(a_1, a_4, a_5)$$

Runtime

The *Klee-Minty cubes* [22] [29] are the polytopes of the linear programs in $\mathbb{R}^n$ with $m = 2n$ facets given by

$$\begin{aligned} \min x_n : \\ 0 \leq x_1 \leq 1, \\ \varepsilon x_{i-1} \leq x_i \leq 1 - \varepsilon x_{i-1} \end{aligned}$$

for $2 \leq i \leq n$ and $0 < \varepsilon < 1/2$. Our illustration shows the 3-dimensional Klee-Minty cube for $\varepsilon = 1/3$.

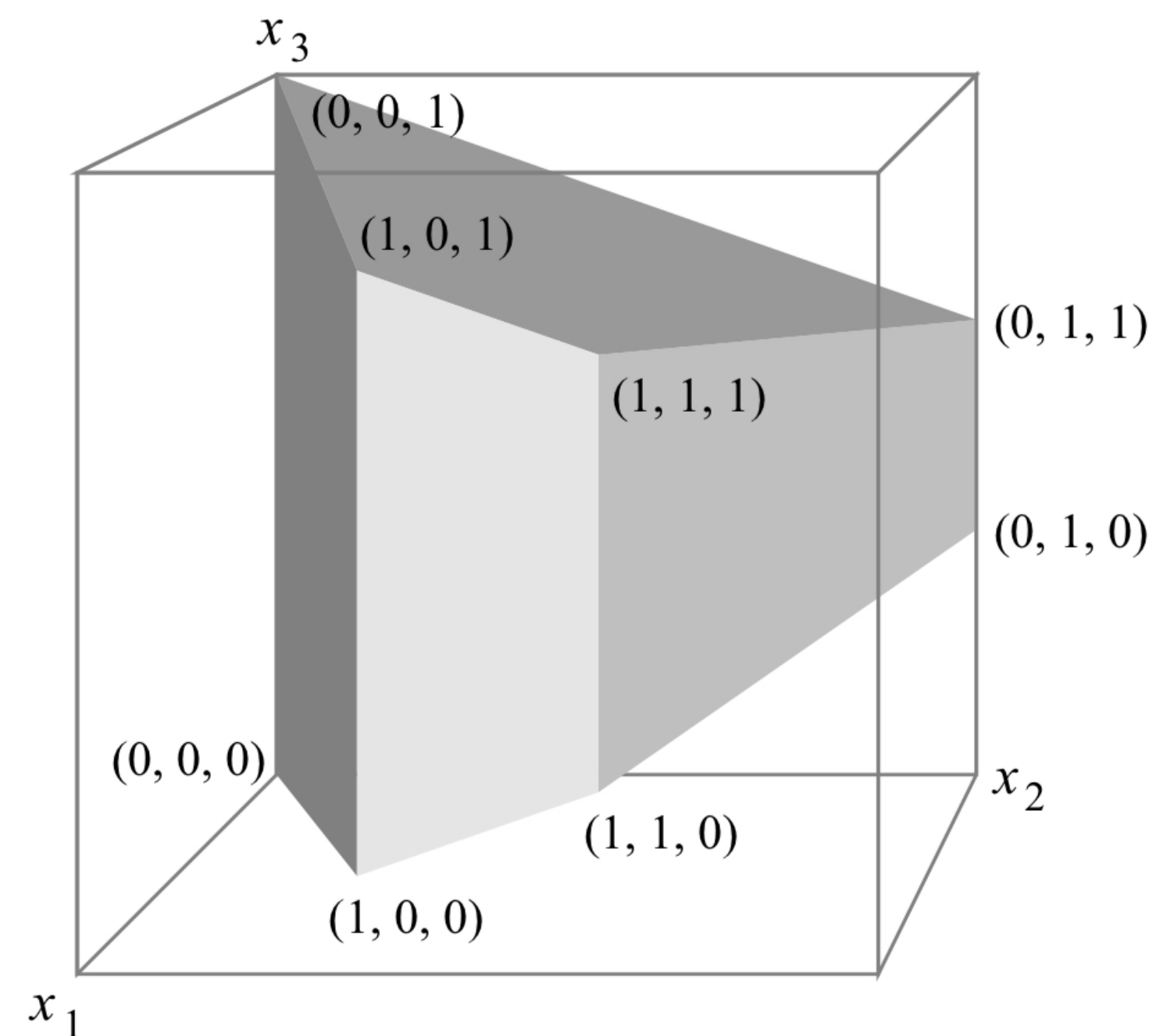


Figure 1: Klee-Minty cube for $n = 3, \varepsilon = 1/3$

Considering the geometry in the limit $\varepsilon \rightarrow 0$, one sees that the feasible region is a (slightly) deformed unit cube.

from :
"Randomized
Simplex
Algorithms
on Klee -
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by
B. Gärtner,
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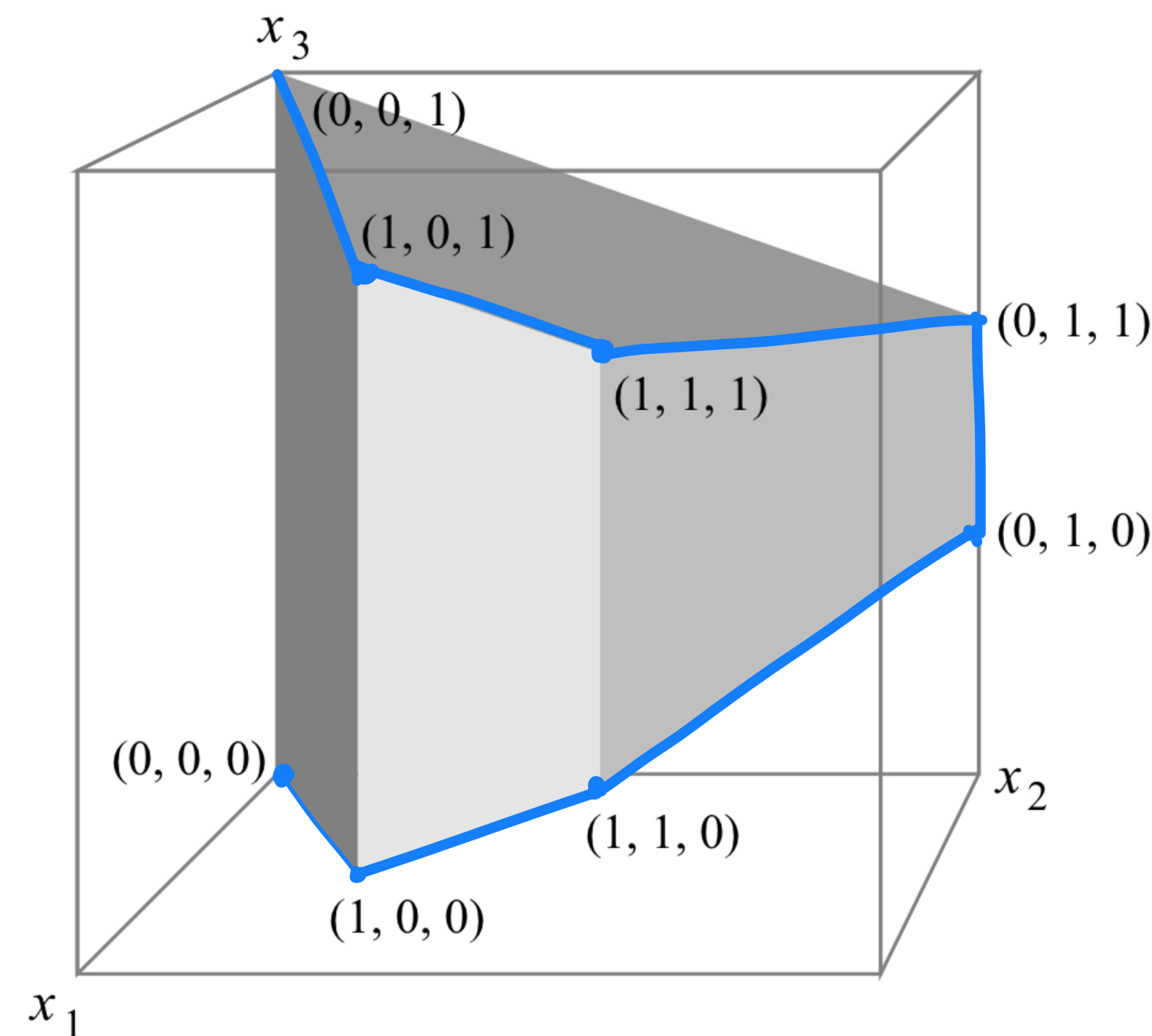


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Can we always "fool" Simplex

like this?

– for linear objectives c unknown

- for linear objectives c unknown
- for quadratic objectives answer is yes

- for linear objectives c unknown
- for quadratic objectives answer is yes:

"An unconditional lower bound for the active-set method in convex quadratic maximization"

joint work with Yann Disser,

Sophie Huiberts and Nils Mosis

Algorithm 1: $\text{SIMPLEX}(\mathbf{c}, A, \mathbf{b}, \mathbf{x})$

input: $\mathbf{c} \in \mathbb{Q}^n$, vertex $\mathbf{x}$ of polytope $\mathcal{P} := \{\mathbf{y}: A\mathbf{y} \leq \mathbf{b}\}$,
system $A\mathbf{y} \leq \mathbf{b}$ non-degenerate

output: optimum solution $\mathbf{x}$ of $\max\{\mathbf{c}^\top \mathbf{y}: \mathbf{y} \in \mathcal{P}\}$

$\mathcal{B} \leftarrow \text{eq}(\mathbf{x})$ (basis)

while $\mathcal{D} := \{\mathbf{d}: A_{\text{eq}(\mathbf{x})}.\mathbf{d} \leq \mathbf{0}, \mathbf{c}^\top \mathbf{d} > 0\} \neq \emptyset$:

take $\mathbf{d} \in \mathcal{D}$ with $|\{i \in \mathcal{B}: A_i.\mathbf{d} = 0\}| = n - 1$

if $A_{\mathcal{B}}.\mathbf{d} \neq \mathbf{0}$: (always true)

$\mathcal{B} \leftarrow \mathcal{B} \setminus \{\ell\}$ for the unique $\ell \in \mathcal{B}$ with $A_\ell.\mathbf{d} < 0$

if $A_{\mathcal{B}}.\mathbf{d} = \mathbf{0}$: (always true)

$\mu \leftarrow \inf\{\mu \geq 0: \mathbf{x} + \mu\mathbf{d} \notin \mathcal{P}\}$

$\mathbf{x} \leftarrow \mathbf{x} + \mu\mathbf{d}$

if $\mathbf{c}^\top \mathbf{d} > 0$: (always true)

$\mathcal{B} \leftarrow \mathcal{B} \cup \{j\}$ for the unique $j \in \text{eq}(\mathbf{x}) \setminus \mathcal{B}$

Algorithm 2: $\text{ACTIVESET}(f, A, \mathbf{b}, \mathbf{x})$

input: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ continuously differentiable,
point $\mathbf{x}$ of polytope $\mathcal{P} := \{\mathbf{y}: A\mathbf{y} \leq \mathbf{b}\}$

output: critical point $\mathbf{x}$ of $\max\{f(\mathbf{y}): \mathbf{y} \in \mathcal{P}\}$

$\mathcal{A} \leftarrow \text{eq}(\mathbf{x})$ (active set)

while $\mathcal{D} := \{\mathbf{d}: A_{\text{eq}(\mathbf{x})}.\mathbf{d} \leq \mathbf{0}, \nabla f(\mathbf{x})^\top \mathbf{d} > 0\} \neq \emptyset$:

take $\mathbf{d} \in \mathcal{D}$ maximizing $|\{i \in \mathcal{A}: A_i.\mathbf{d} = 0\}|$

if $A_{\mathcal{A}}.\mathbf{d} \neq \mathbf{0}$:

$\mathcal{A} \leftarrow \mathcal{A} \setminus \{i\}$ for some $i \in \mathcal{A}$ with $A_i.\mathbf{d} < 0$

if $A_{\mathcal{A}}.\mathbf{d} = \mathbf{0}$:

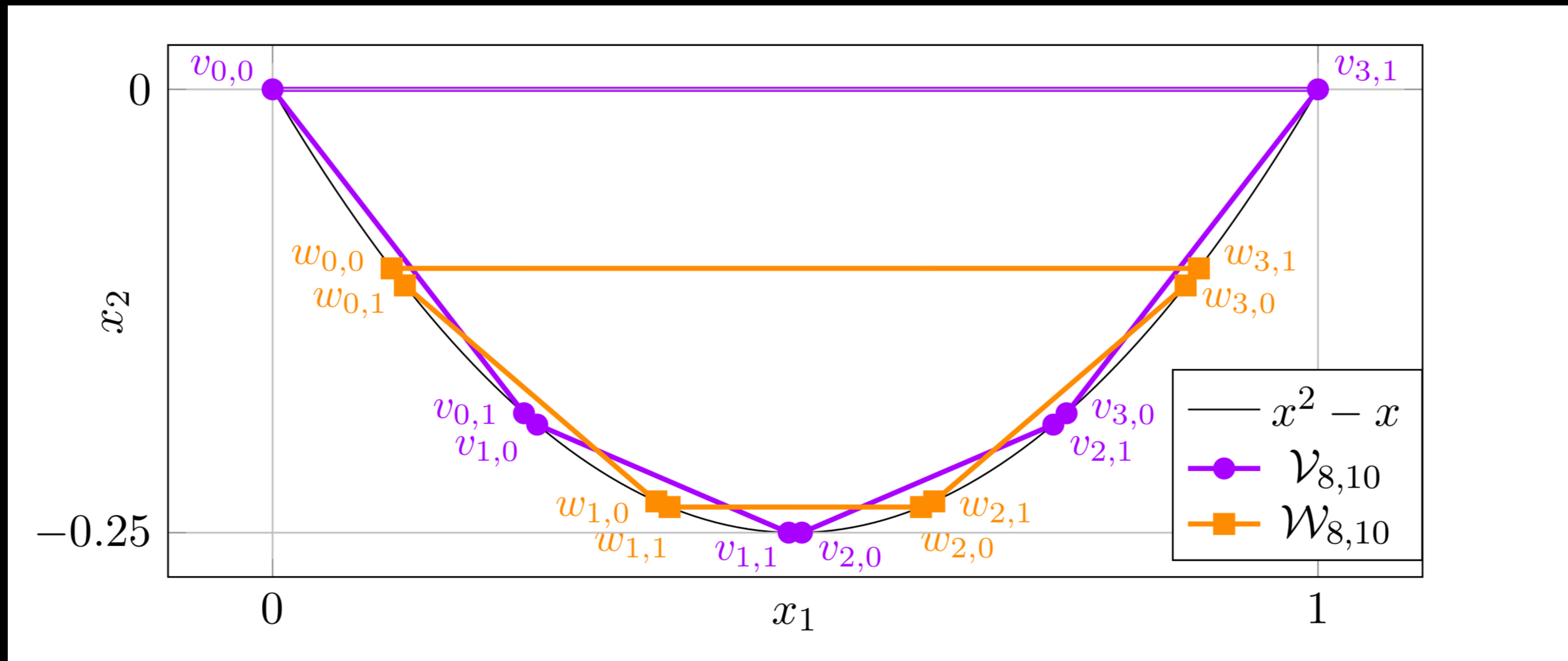
$\mu \leftarrow \inf\{\mu \geq 0: \mathbf{x} + \mu\mathbf{d} \notin \mathcal{P} \text{ or } \nabla f(\mathbf{x} + \mu\mathbf{d})^\top \mathbf{d} \leq 0\}$

$\mathbf{x} \leftarrow \mathbf{x} + \mu\mathbf{d}$

if $\nabla f(\mathbf{x})^\top \mathbf{d} > 0$:

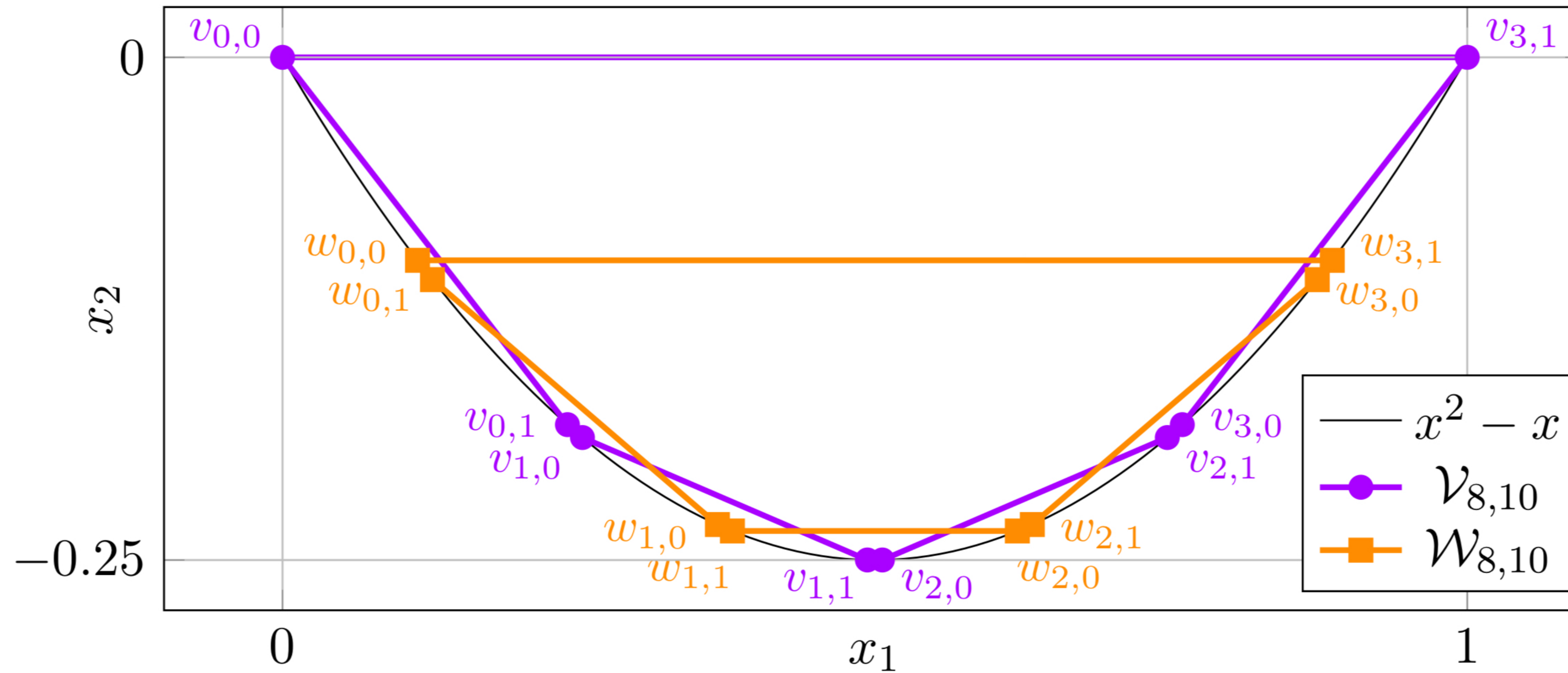
$\mathcal{A} \leftarrow \mathcal{A} \cup \{j\}$ for some $j \in \text{eq}(\mathbf{x}) \setminus \mathcal{A}$

Disser b Mosis' 25



– extended formulation via deformed products
(Amenta & Ziegler '99)

$$f((x_1, x_2)) = x_1^2 - x_1 - x_2$$



$$f((x_1, x_2)) = x_1^2 - 2x_1 - x_2$$

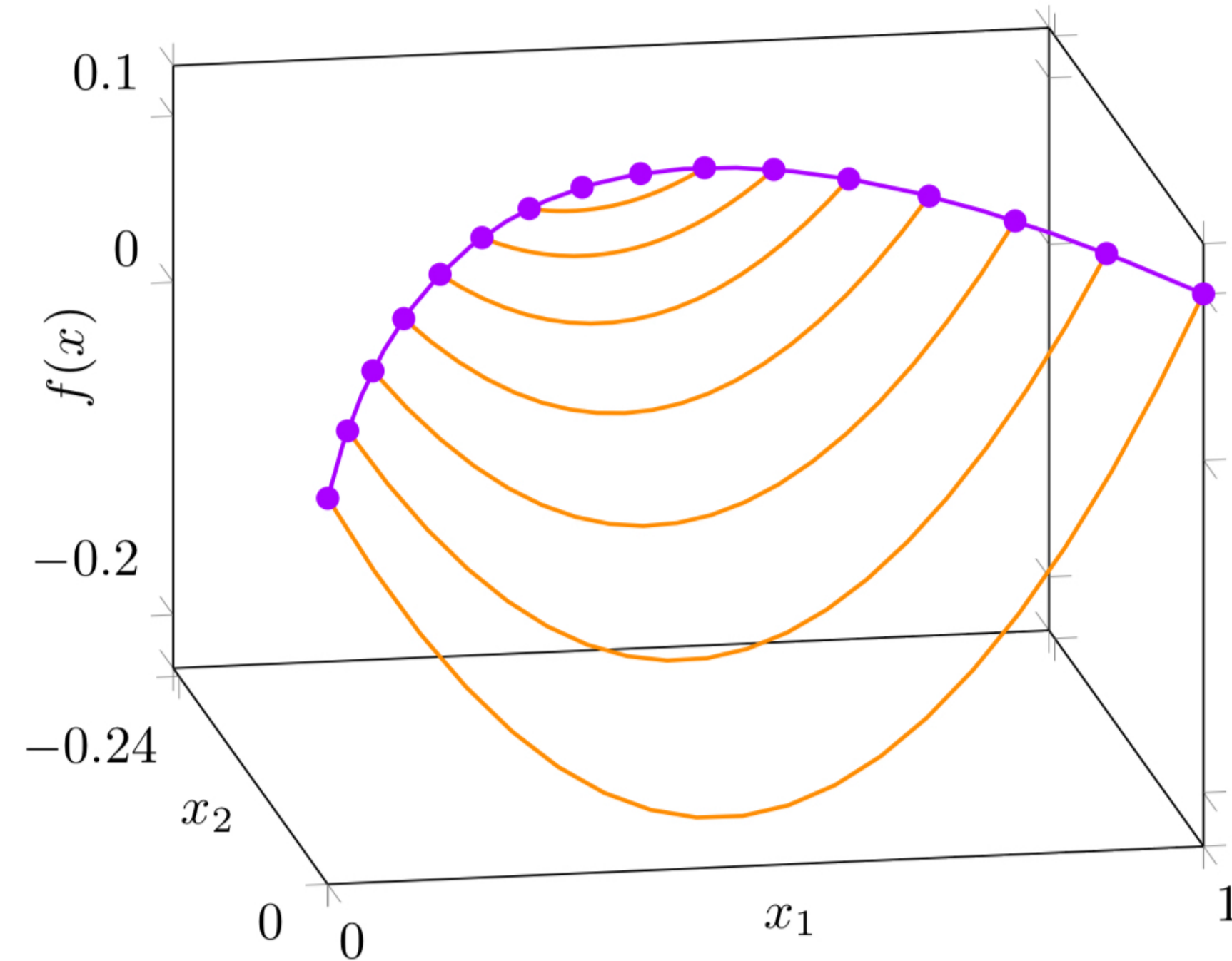


Figure 4: Objective function values along vertices and edges of the projection $\Pi(\mathcal{P})$ in (QP) for $n = M = 16$, $d = 4$. Purple edges are monotonically increasing in objective value, while the orange edge and chords have negative gradients at both endpoints.

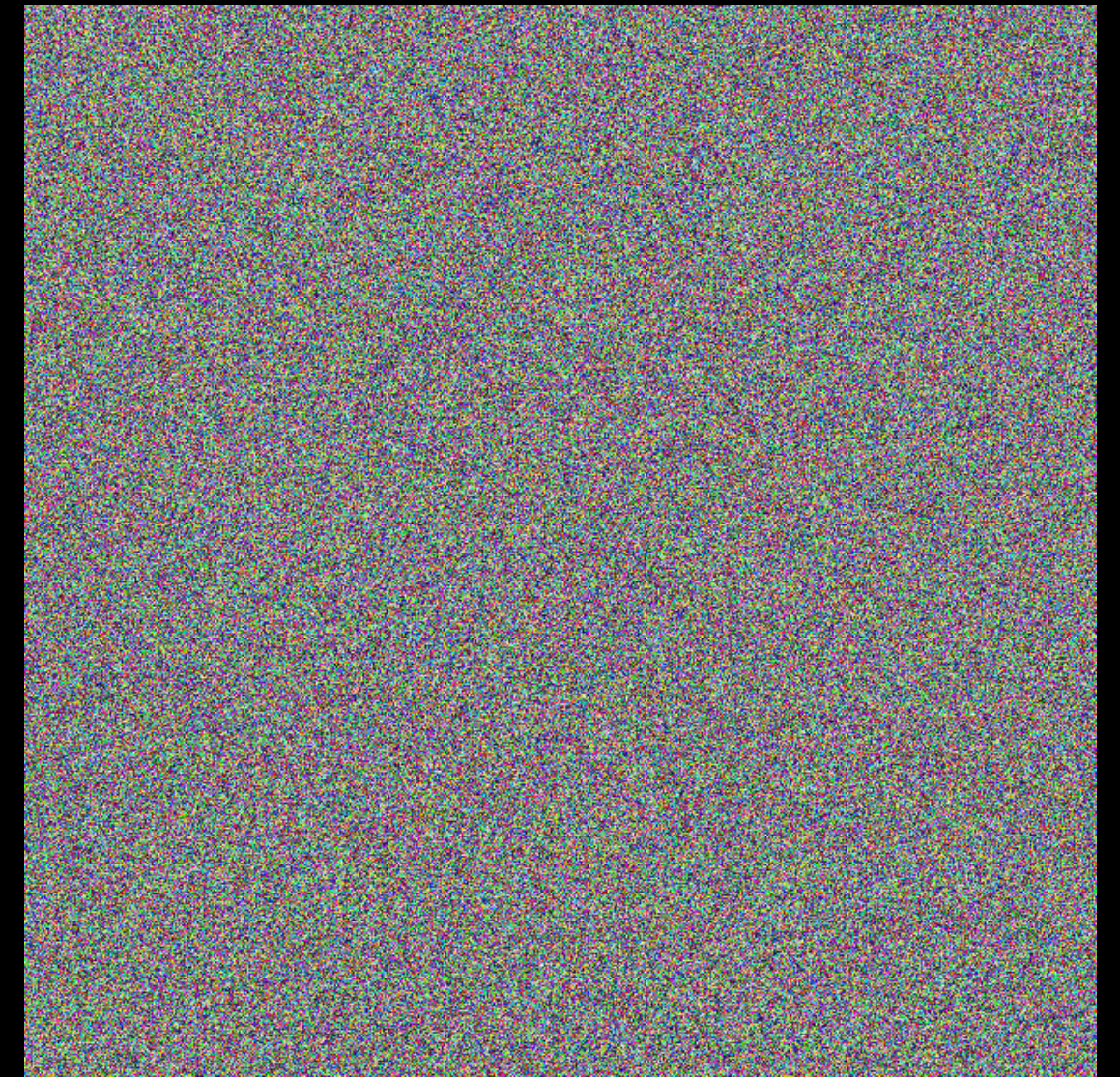
Practice: run time linear in n and d .

How to explain / model this?

Worst case:



Average case:



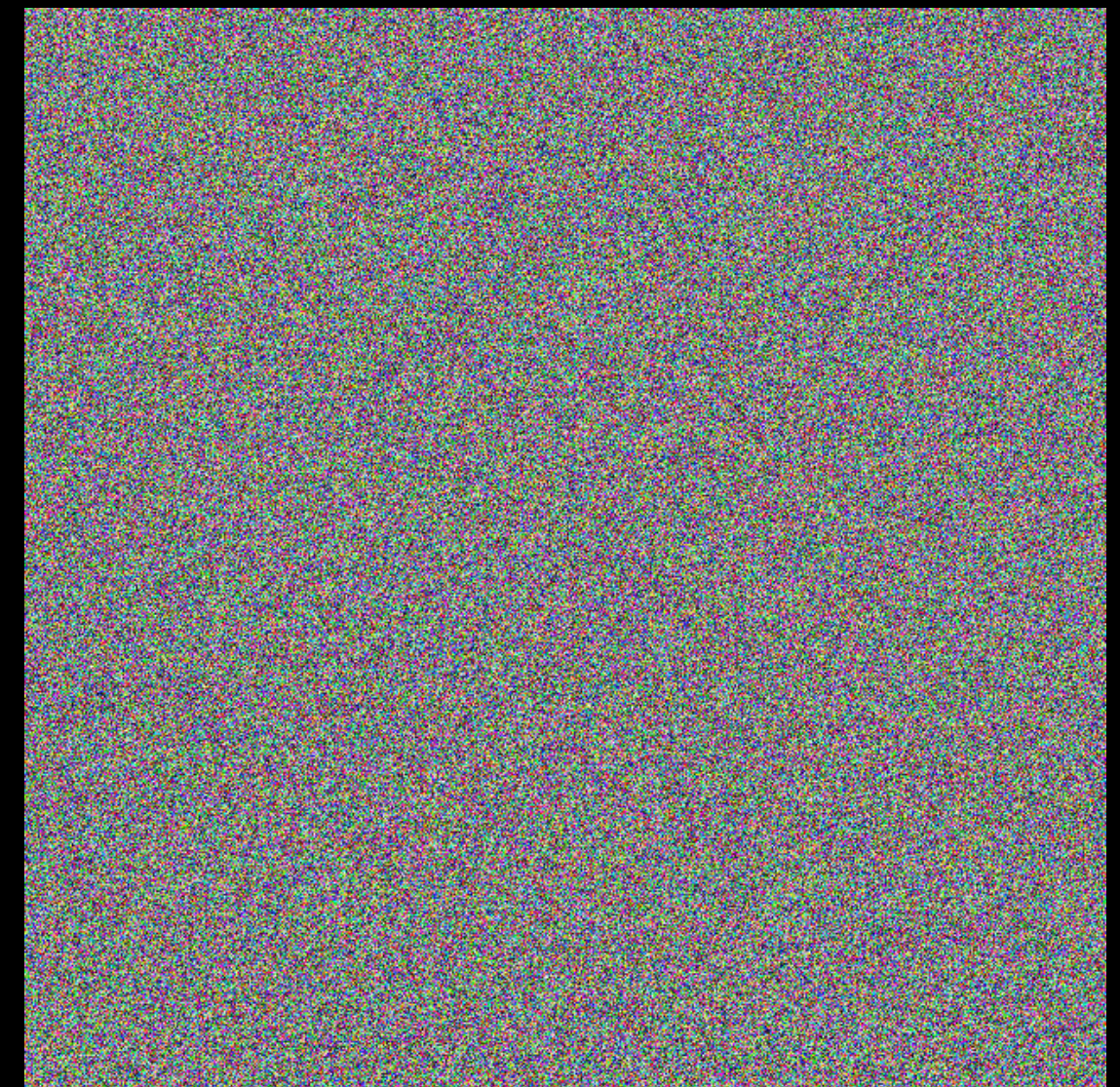
Worst case:
Structure ✓

Upper bounds
(poly) ✗



Average case:
Structure ✗

Upper bounds
(poly) ✓



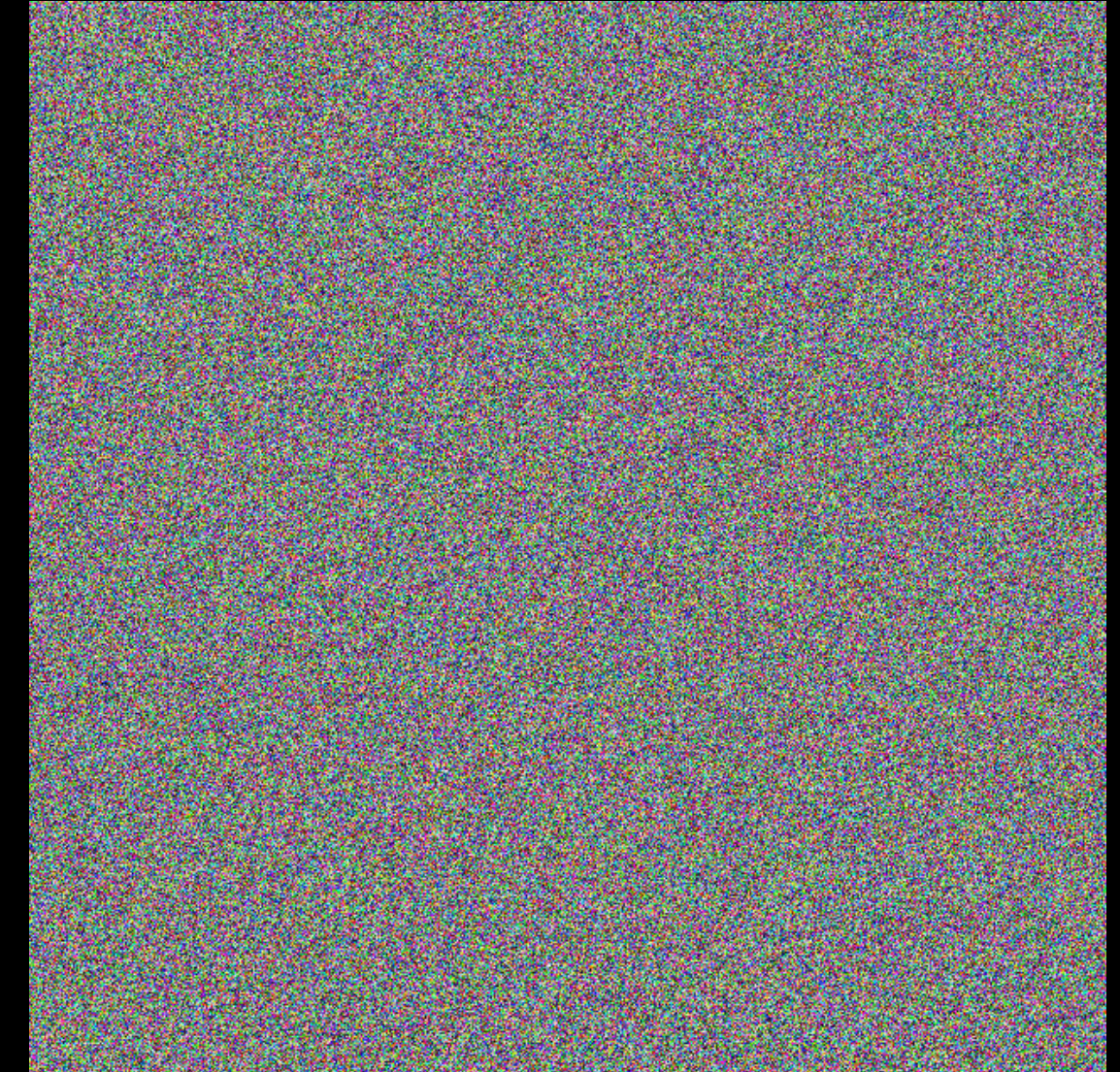
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Worst case:

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Upper bounds
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Smoothed analysis.

Structure ✓

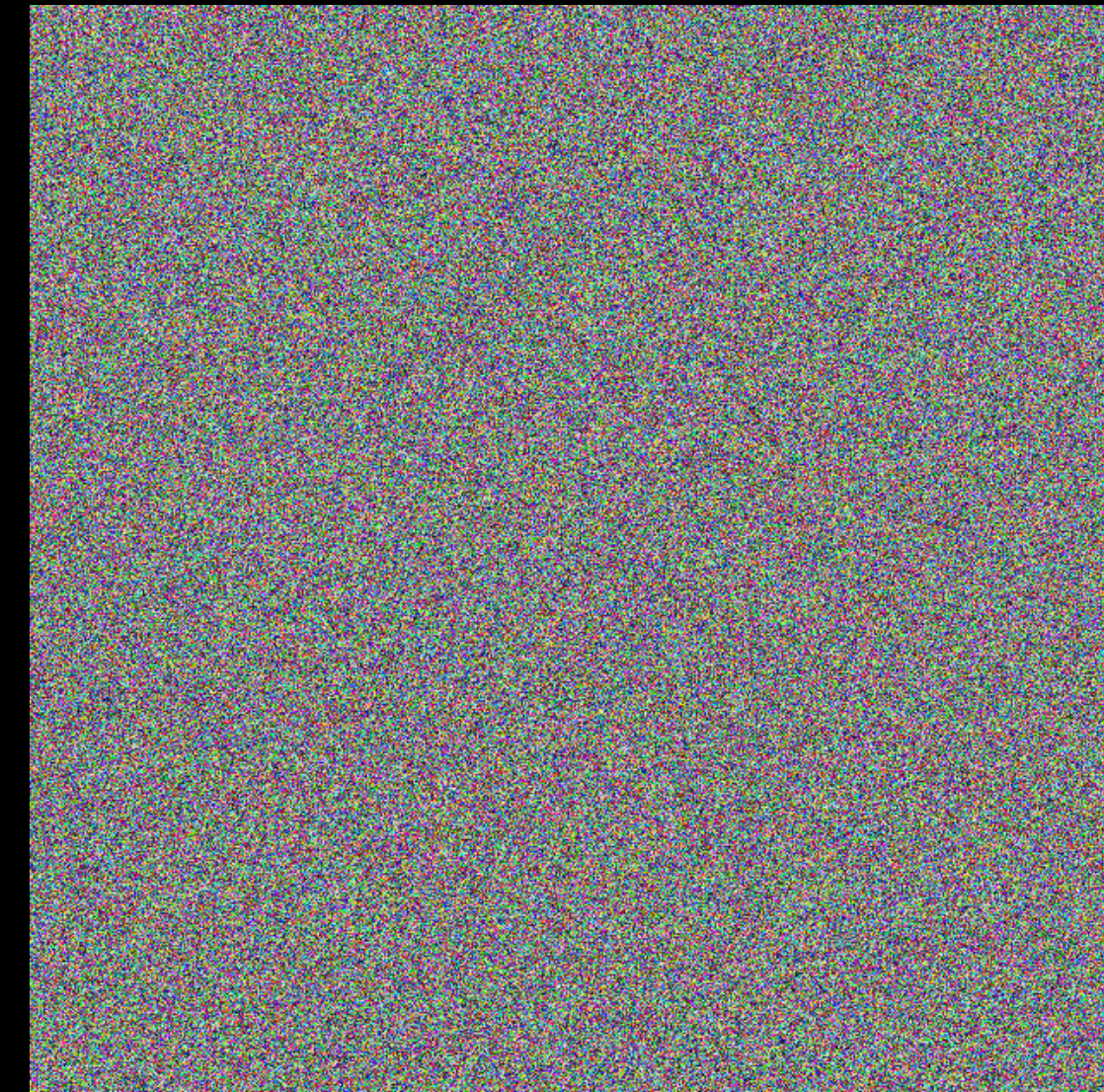
Upper bounds (poly) ✓



Average case:

Structure ✗

Upper bounds
(poly) ✓



Smoothed analysis

Let $\bar{A} \in \mathbb{R}^{n \times d}$ have rows of norm ≤ 1 .

$\bar{b} \in [-1, 1]^n$, $c \in \mathbb{R}^d$

Let $\hat{A}, \hat{b}$ have i.i.d. $N(0, \sigma^2)$ entries

Smoothed analysis

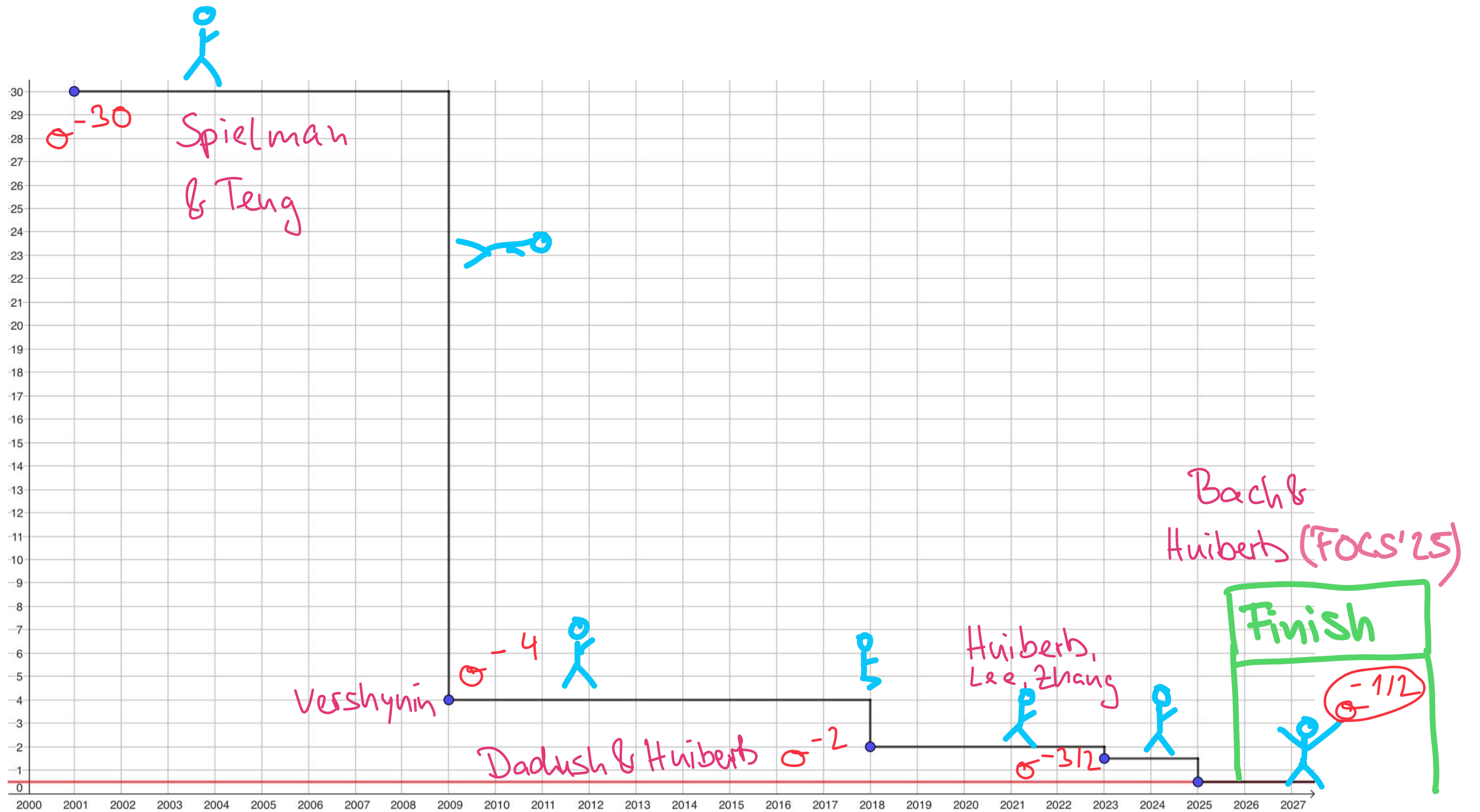
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Theorem:

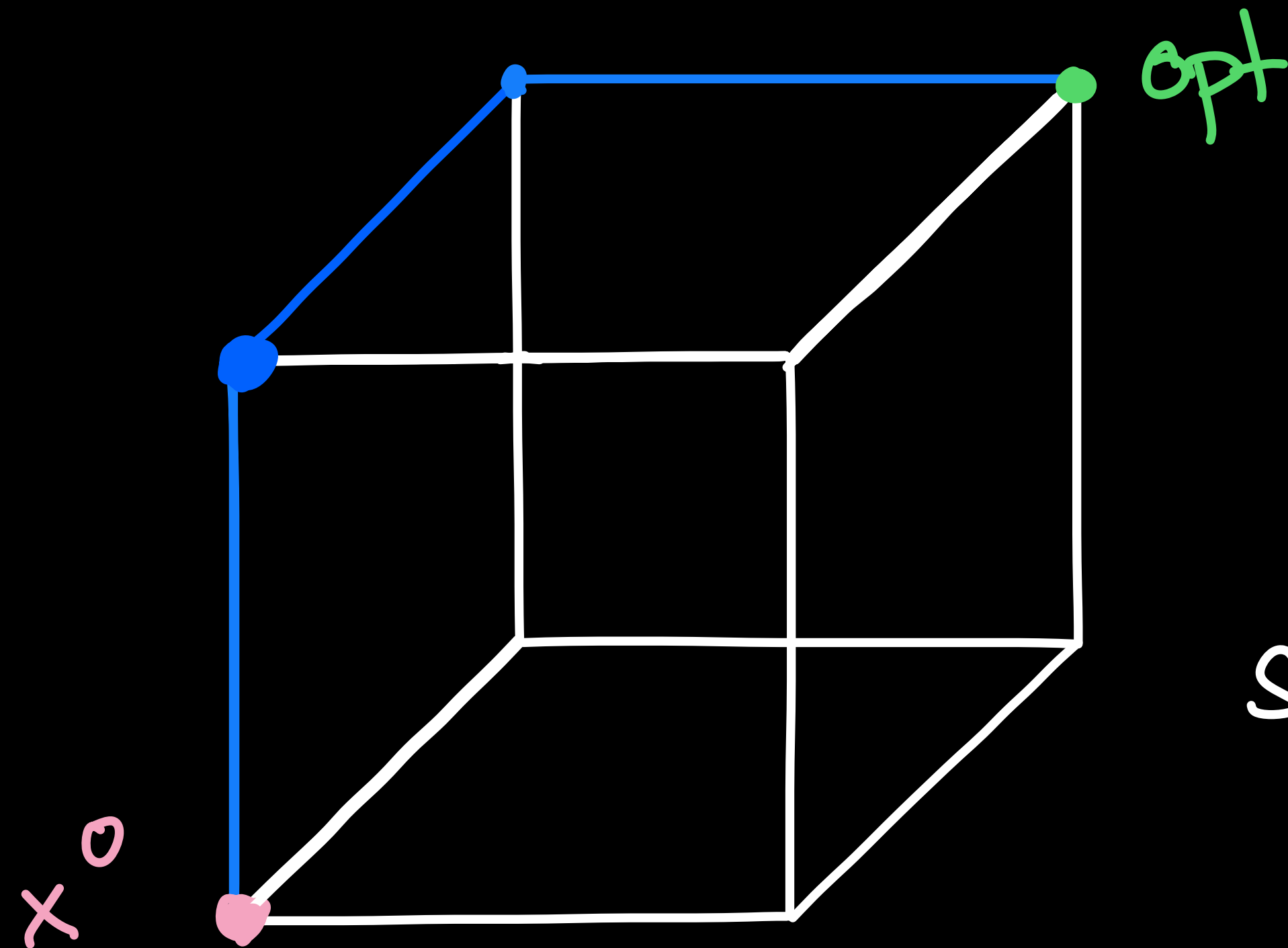
$$\max_{\bar{A}, \bar{b}, c} \mathbb{E}_{\hat{A}, \hat{b}} \left[\text{time to solve} \begin{array}{l} \max c^T x \\ \text{s.t. } (\bar{A} + \hat{A})x \leq \bar{b} + \hat{b} \end{array} \right] = \sqrt{\sigma^{-1}} \cdot \text{poly}(d, \log n)$$



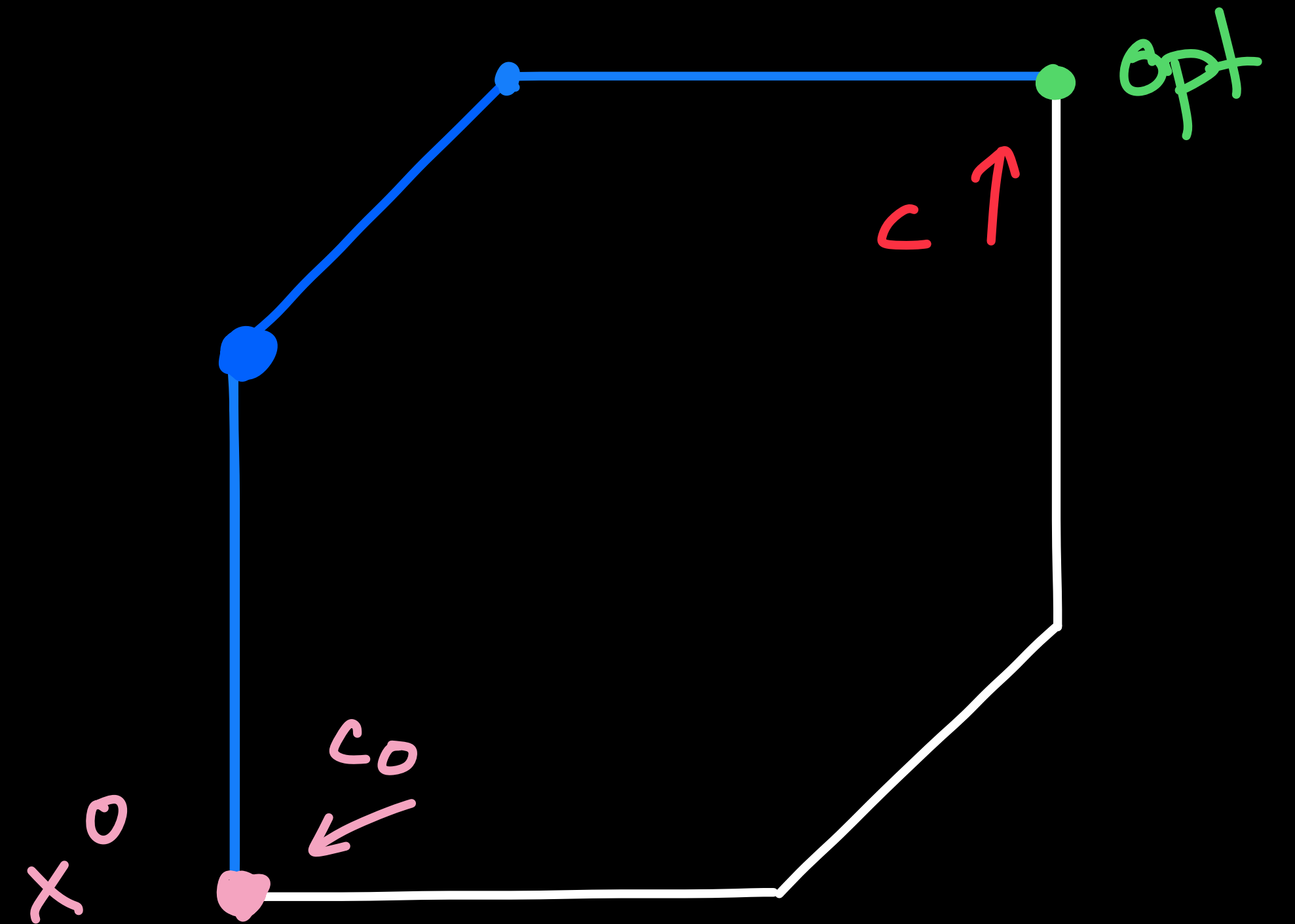
Shadow vertex

Simplex

method



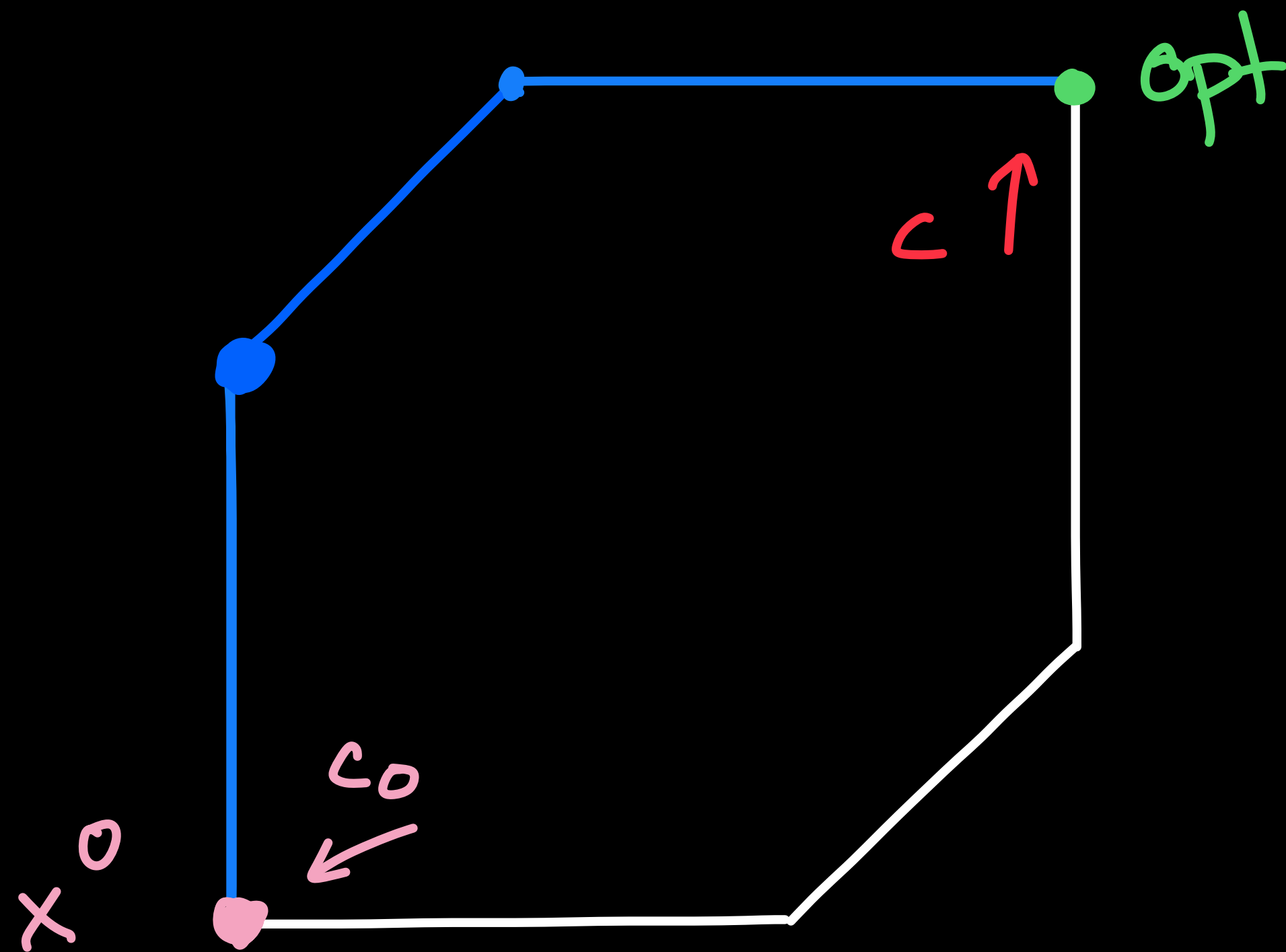
project to
→
 $\text{span}(c_0, c)$



Shadow vertex

Simplex

method



- solutions to

$$\max (\lambda c_0 + (1-\lambda)c)^T x$$

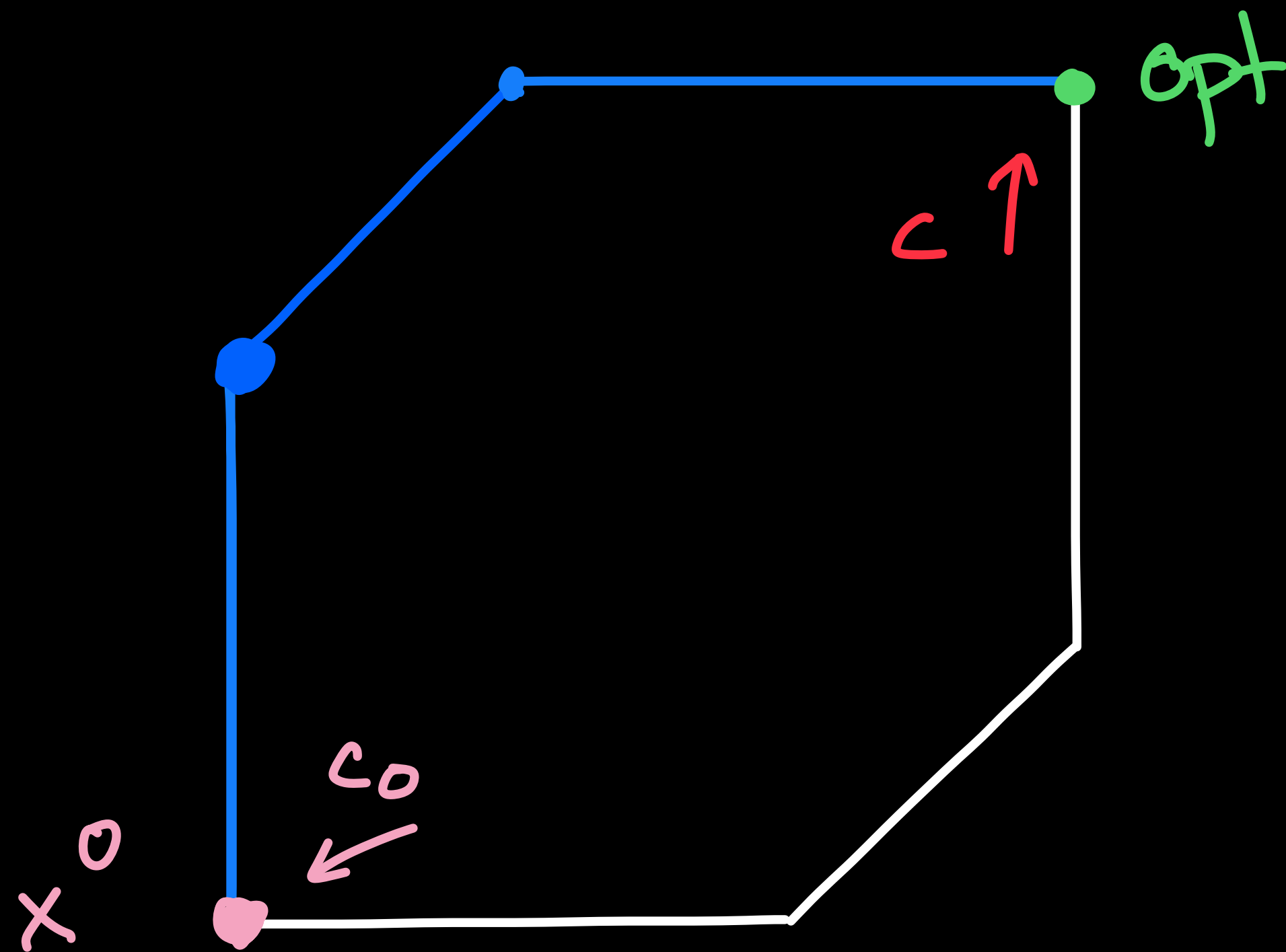
$$\text{s.t. } Ax \leq b$$

form simplex path

Shadow vertex

Simplex

method



- solutions to

$$\max (\lambda c_0 + (1-\lambda)c)^T x$$

$$\text{s.t. } Ax \leq b$$

form simplex path

→ we bound # vertices on path

How to bound # vertices?

1. Choose quantities to measure on $\mathbb{R}^d$ -polytope

2. project $\{x: Ax \leq b\}$ to $\text{span}(c_0, c)$

3. Choose potentials

perimeter

polar perimeter

etc.

exterior angle

polar exterior angle

4. potential measure

Our improvements

1. Slack ($Ax \leq b \rightarrow Ax + s = b$), multipliers (normal cone $\mathcal{N}(x) = \{y \in \mathbb{R}^d : y = \sum \lambda_i a_i, \lambda_i \geq 0\} = \{y \in \mathbb{R}^d : A_x^{-1} y \geq 0\}$)
 $\geq m$

Our improvements

1. slack ($Ax \leq b \rightarrow Ax + s = b$), multipliers (normal cone

$$N(x) = \{y \in \mathbb{R}^d : y = \sum \lambda_i a_i, \lambda_i \geq 0\} = \{y \in \mathbb{R}^d : A_x^{-1} y \geq 0\})$$

2. pick c_0 random, c fixed (apart algo reduction)



independent of $\bar{A}, \bar{b}$ (and c)

Our improvements

1. slack ($Ax \leq b \rightarrow Ax + s = b$), multipliers (normal cone $N(x) = \{y \in \mathbb{R}^d: y = \sum \lambda_i a_i, \lambda_i \geq 0\} = \{y \in \mathbb{R}^d: A_x^{-1} y \geq 0\}$)
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3. pick two clever potentials
"perimeter integral" & "exterior angle"

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2. pick c_0 random, c fixed (apart algo reduction)
3. pick two clever potentials
"perimeter integral" & "exterior angle"
4. combine them to vertex-neighbor separation
 $\rightarrow$ each step consumes a lot of either integral

Does smoothed analysis

explain efficiency of the

Simplex method?

Problem 1:

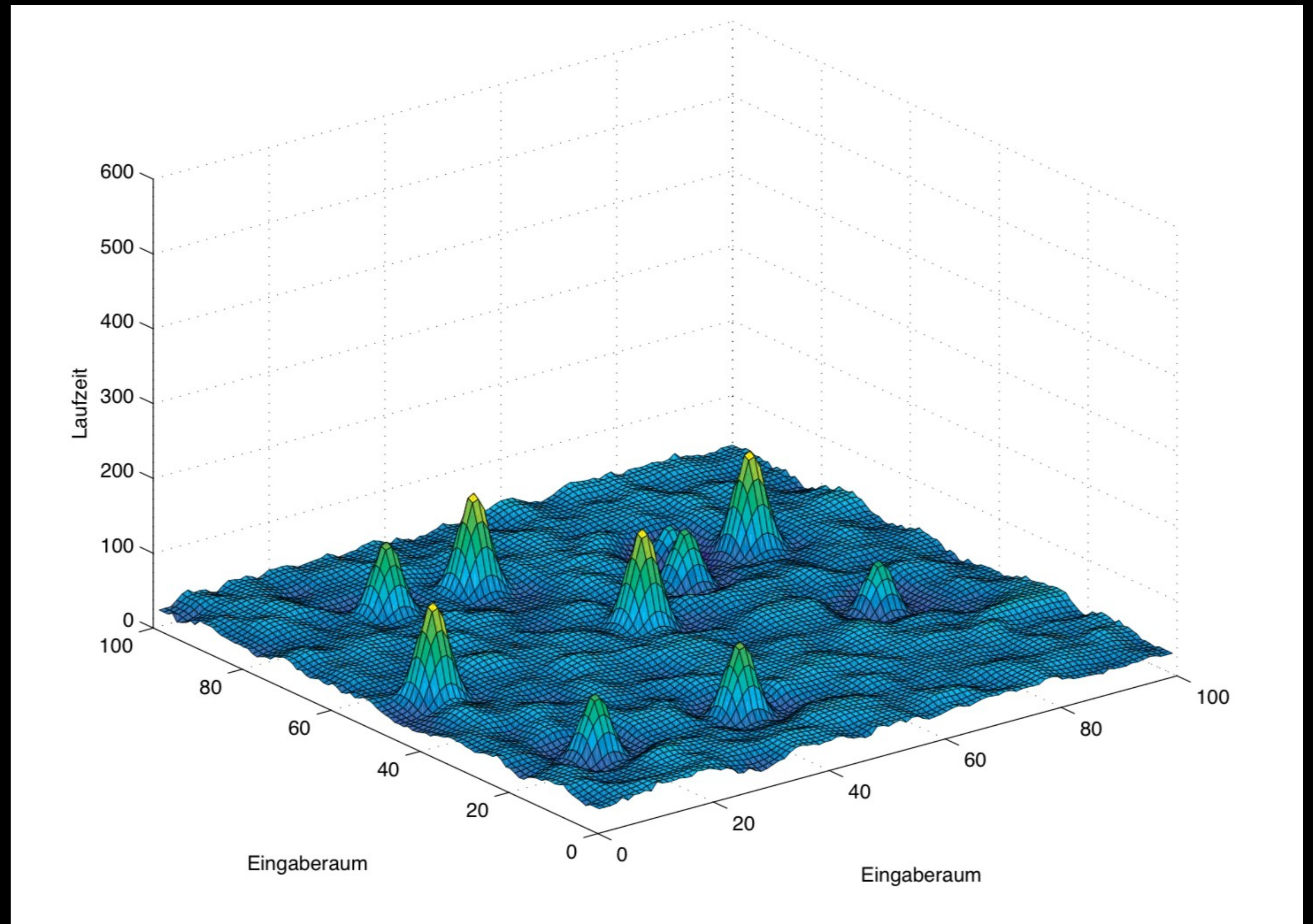
Real world LP:

$< 0.1\%$ of $A_{i,j} \neq 0$

Smoothed analysis

all $A_{i,j} \neq 0$

Problem 2:



K. H. Borgwardt: "Wie schnell arbeitet das Simplexverfahren normalerweise?"

Problem 3:

What are we modelling? What is σ ?

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- process within solver? $\rightarrow 10^{-6}$ solver tolerance?

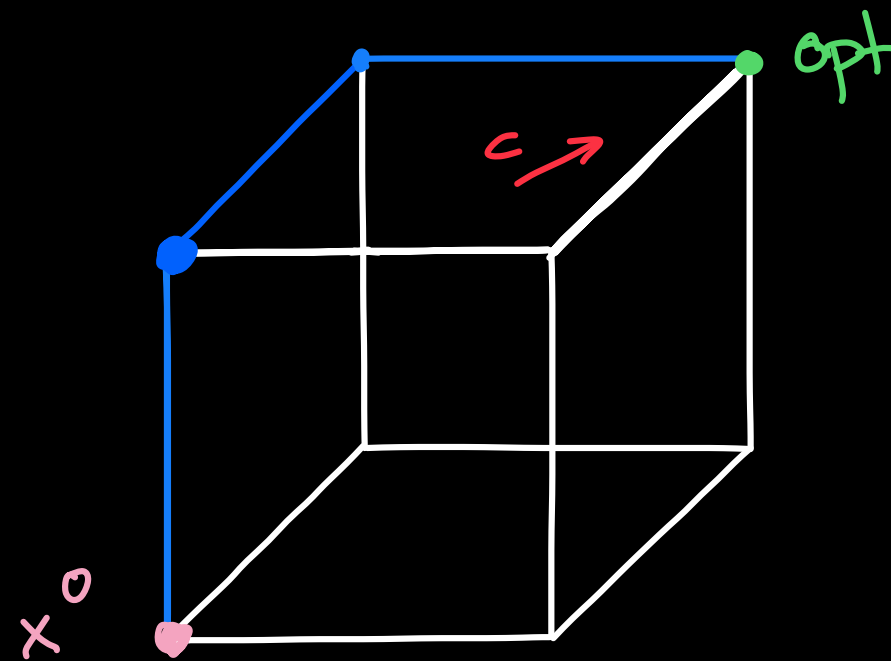
Problem 3:

What are we modelling? What is σ ?

- measurement error? $\rightarrow 10^{-2}$ sigfigs accuracy
- brittleness?
- process within solver? $\rightarrow 10^{-6}$ solver tolerance?
- floating point imprecision? 10^{-16} 64-bit IEEE 754?

Are we studying the simplex method?

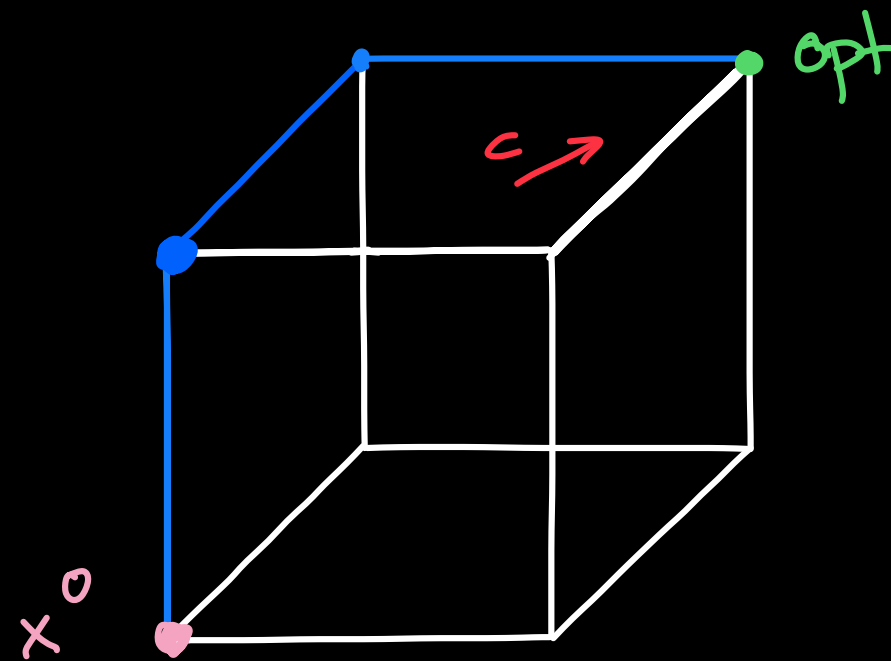
- This geometry



does not exist

Are we studying the simplex method?

- This geometry



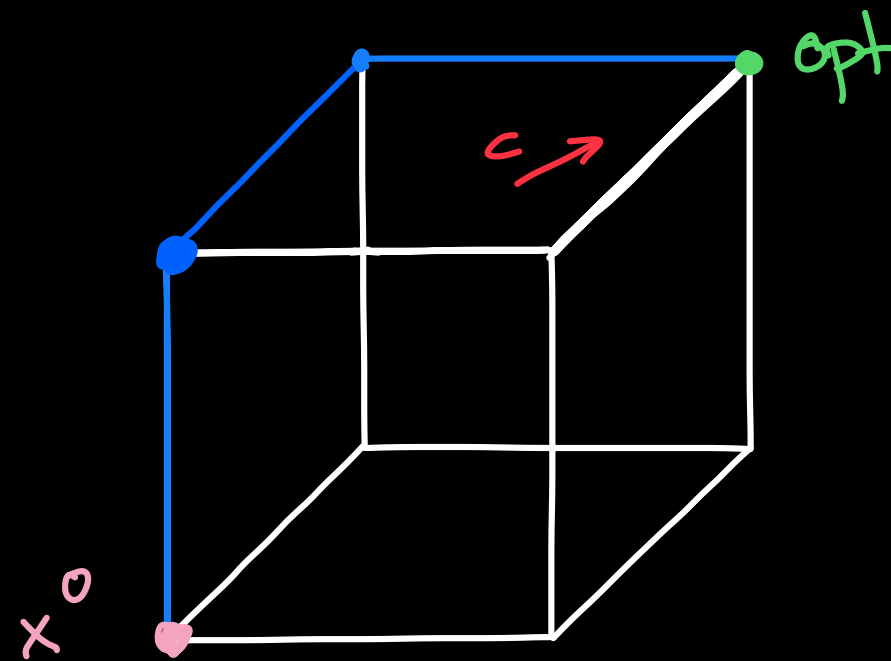
does not exist

- What exists :

{ numerical linear algebra

Are we studying the simplex method?

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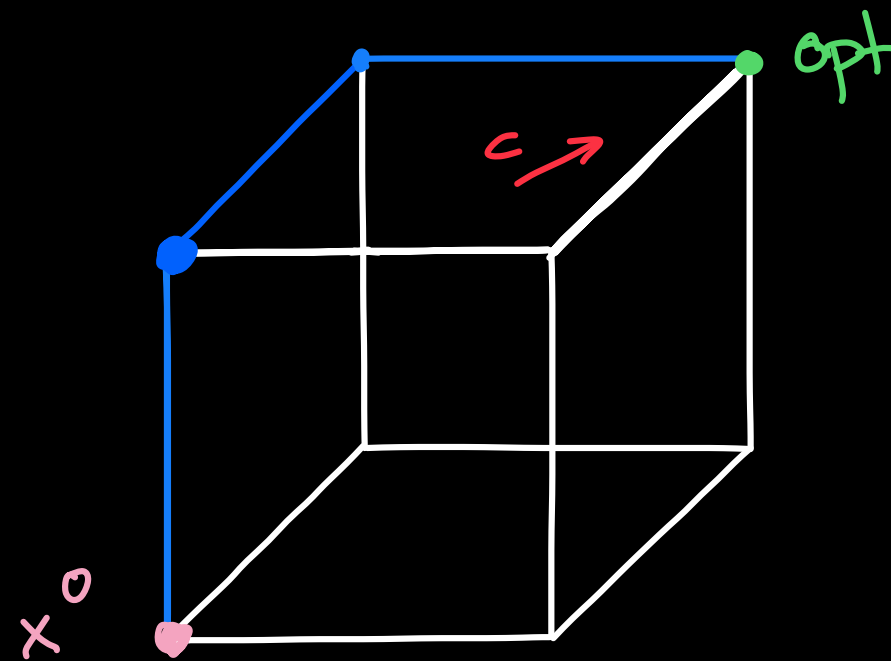
does not exist

- What exists :

{ numerical linear algebra
bound shifting

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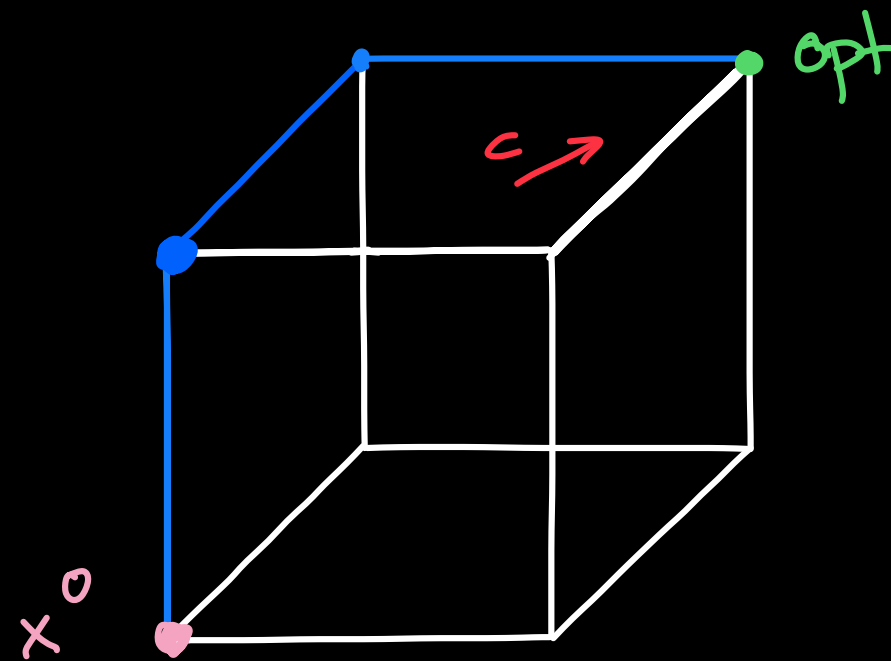
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bound shifting
bound perturbations

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{ numerical linear algebra
bound shifting
bound perturbations
(Harris) ratio test